

CHAPTER 1

INTRODUCTION

1.1. Background

Thunderstorms are among the most dynamic and hazardous phenomena in the atmospheric boundary layer, arising from the rapid vertical transport of moist, unstable air under favourable thermodynamic and kinematic conditions. Thunderstorms remain notoriously difficult to anticipate with the lead time required for effective response, particularly in tropical and sub-tropical regions such as the Indian subcontinent.

Upper air meteorological data, obtained primarily through radiosonde ascents, provide direct measurements of temperature, dew point, wind speed, and wind direction at successive pressure levels through the troposphere. These profiles allow derived indices such as Convective Available Potential Energy (CAPE), Lifted Index, K-Index, and wind shear to be computed, collectively describing the atmosphere's readiness to support deep convection.

1.2. Importance of Thunderstorm Forecasting

Accurate and timely thunderstorm forecasts carry direct implications for public safety, aviation operations, agriculture, and infrastructure protection. Severe thunderstorms can produce cloud-to-ground lightning, damaging wind gusts, and sudden downpours capable of disrupting power supply, transportation, and outdoor activity within a matter of minutes. For sectors such as aviation and construction, even a short-range warning of one to a few hours can be sufficient to reschedule operations and prevent loss of life or equipment damage. At a broader scale, reliable convective forecasts support disaster management agencies in issuing localized alerts, thereby reducing the human and economic cost associated with severe weather events. The value of a thunderstorm forecast, therefore, lies not merely in predicting occurrence but in doing so with sufficient lead time and spatial precision to be operationally actionable.

1.3. Role of Machine Learning in Weather Prediction

Machine learning offers an alternative, data-driven pathway to weather prediction by learning statistical relationships between input meteorological variables and observed outcomes, without requiring an explicit physical formulation of every governing process. When applied to upper air data, machine learning models can simultaneously account for the combined influence of thermodynamic instability, moisture content, and vertical wind structure, capturing interactions that are difficult to encode through fixed empirical thresholds.

1.4. Objectives

The specific objectives of this project are as follows:

1. To compute relevant atmospheric stability indices (such as CAPE, Lifted Index, and K-Index) from the raw radiosonde profiles.
2. To design and implement machine learning models capable of classifying thunderstorm-favourable and thunderstorm-unfavourable atmospheric conditions.
3. To evaluate and compare the performance of different machine learning algorithms using standard classification metrics.
4. To assess the relative importance of individual upper air parameters in driving thunderstorm occurrence.

CHAPTER 2

DATASET DESCRIPTION

2.1. Overview

The dataset used in this project spans twenty-five years of upper-air meteorological observations (2000–2025) recorded at station 43150 (Visakhapatnam, India; 17.70° N, 83.30° E). It combines two independently sourced records for the same station and period: radiosonde-derived atmospheric soundings, from which stability and moisture indices were computed, and surface SYNOP present-weather reports, from which the thunderstorm occurrence label was derived. The resulting tabular dataset associates, for each observation time, a vector of upper air-derived features with a binary indicator of whether a thunderstorm was reported within the following 24 hours.

2.2. Data Source

Two primary data sources were used. The first is the archive of upper air soundings for station 43150, publicly available through the University of Wyoming radiosonde database, which records the full vertical profile of pressure, height, temperature, dewpoint, relative humidity, mixing ratio, wind direction, and wind speed at each mandatory and significant level of the atmosphere, along with a station header giving the observation time, latitude, longitude, and elevation. The second is the SYNOP surface observation archive for the same station, from which present-weather (ww) and past-weather (W1, W2) codes were extracted to construct the ground-truth thunderstorm label. Both sources were aligned by date and observation hour so that each sounding could be matched to the corresponding surface report.

2.3. Upper Air Sounding and Derived Indices

An upper air sounding is the vertical profile obtained from a single radiosonde ascent, typically visualised on a thermodynamic (skew-T log-P) diagram. Rather than using the raw multi-level profile directly as model input, each sounding was condensed into a fixed-length feature vector of stability, moisture, and shear indices computed from the profile using the MetPy atmospheric-science library

ID	Date	Time	SI	LI	U-V	SWAT	CT	VT	TT	CAPE	CAPE_V	CIN	CIN_V	BRN	BRN_V	LCL_T	LCL_P	THETA_L	U_MSL	PT	MMR_MSL	THK_1000_PW	EL	EL_VT	UFC	UFC_V	KI	Thunderstorm_24h		
1	01-01-2001	00Z	5.63	4.37	3.99	170.82	17.2	15.5	36.7	0	0	0	-31	0	0	288.22	906.18	331.34	296.46	12.04	5787	29.14		851.15	851.48		10.4	0		
2	02-01-2001	00Z	7.31	3.95	3.88	51.99	10.5	26.5	37	0	0	0	0	0	0	286.74	897.65	327.65	295.74	11.04	5757	31.85					11.5	0		
3	03-01-2001	00Z	10.2	3.64	3.68	46.02	5.7	25.7	31.4	0	0	0	0	0	0	288.52	911.29	333.63	296.29	12.22	5756	30.95					7.5	0		
4	04-01-2001	00Z	12.78	4.89	4.77	34.01	2.7	23.7	26.4	0	0	0	0	0	0	288.81	918.95	331.54	295.89	12.35	5756	29.11					0.7	0		
5	05-01-2001	00Z	1.3	4.61	4.37	195.21	18.8	23.5	42.3	0	0	0	0	0	0	289.67	920.59	334.33	296.62	13.04	5754	48.1					35.7	0		
6	06-01-2001	00Z	0.14	3.32	3.37	213.19	20.2	23.5	43.7	0	0	0	0	0	0	290.6	929.67	336.36	296.73	13.69	5776	51.19					34.2	0		
7	07-01-2001	00Z	5.88	1.62	1.56	121.62	12.5	25.5	38	2.59	2.87	-435.75	-384.61	0.21	0.23	290.96	922	338.92	297.81	14.14	5773	44.39	446.16	446.33	577.6	579.17	27.8	0		
8	08-01-2001	00Z	7.67	1.2	1.08	104.59	10.1	25.1	35.2	0	0	0	0	0	0	292.39	945.06	340.95	297.18	15.1	5791	42.85					22.9	0		
9	09-01-2001	00Z	7.7	3.99	3.87	87.19	10.7	24.7	35.4	0	0	0	0	0	0	287.88	889.83	332.6	297.66	12.16	5780	32.45					13.3	0		
10	10-01-2001	00Z	7.21	7.39	7.28	118.58	13.1	22.1	35.2	0	0	0	0	0	0	283.41	829.28	327.13	299	9.95	5766	32.46					14.9	0		
11	11-01-2001	00Z	3.43	3.56	3.34	163	16.9	22.9	39.8	0	0	0	0	0	0	288.7	884.74	336.23	299.01	12.93	5778	42.56					29.9	0		
12	12-01-2001	00Z	4.5	9.04	8.87	174.2	16.3	20.9	37.2	0	0	0	0	0	0	283.08	822.83	327.03	299.3	9.44	5787	35.35					23.5	0		
13	13-01-2001	00Z	7.96	5.35	4.76	160.98	13.1	17.7	30.8	6.46	13.88	-75.33	-52.54	2.71	5.82	292.11	937.2	341.06	297.59	14.98	5774	36.54	781.8	780.12	815.83	828.96	17.7	0		
14	14-01-2001	00Z	6.11	6.93	6.72	193.8	16.3	17.1	33.4	0	0.43	-50.93	-42.28	0	0.05	287.2	870.67	333.14	298.82	11.86	5770	35.62	806.73	805.57	811.38	821.21	15.5	0		
15	15-01-2001	00Z	5.98	4.1	3.71	175.38	16.1	17.7	33.8	9.57	17.06	-20.36	-10.53	2.59	4.62	291.22	923.45	339.69	297.94	14.39	5789	36.86	811.35	790.48	859.49	876.37	12.9	0		
16	16-01-2001	00Z	7.92	3.89	3.49	156.41	13.7	18.7	32.4	0.74	5.04	-37.87	-27.09	0.48	3.31	291.33	932.1	338.78	297.26	14.32	5786	35.56	824.15	823.19	840.8	852.86	11.9	0		
17	17-01-2001	00Z	6.15	2.97	2.53	148.2	15.3	20.3	35.6	0	0	0	0	0	0	291.06	929.75	338.12	297.21	14.13	5781	37.41					19.9	0		
18	18-01-2001	00Z	4.58	4.69	4.4	179.8	17.6	19.3	36.9	0	0.17	0	-48.39	0	0.02	288.96	904.83	334.14	297.36	12.69	5787	37.32					13.2	0		
19	19-01-2001	00Z	17.55	5.25	4.9	37.63	1.37	16.86	18.13	32.27	84.42	0	0	0	0	941	25.13	289.74	910.74	336.04	297.59	13.23	5745	26.62	836.76	799.35	910.74	910.74	-2.84	0
20	20-01-2001	00Z	6.64	4.91	4.43	163.99	14.1	20.1	34.2	2.26	6.7	-139.11	-112.9	0.26	0.76	290.58	922.9	337.31	297.33	13.8	5792	34.3	779.63	774.39	799.15	806.93	13.1	0		
21	21-01-2001	00Z	3.44	3.66	3.31	200.19	17.6	21.3	38.9	8.46	12.06	-133.57	-121.23	0.41	0.59	289.08	889.21	336.94	298.98	13.03	5803	40.53	762.32	757.4	796.83	801.86	20.4	0		
22	22-01-2001	00Z	2.76	1.57	1.08	200.8	18.7	21.3	40	58.96	73.4	-45.77	-52.3	4.87	6.06	291.42	926.26	340.02	297.68	14.52	5780	41.17	756.66	733.12	828.77	834.6	17.7	0		
23	23-01-2001	00Z	4.46	0.31	-0.23	161.4	17	21.1	38.1	62.69	104.7	-14.01	-5.74	22.5	37.57	292.65	937.65	343.15	296.11	15.51	5782	38.55	419.52	409.44	851.95	878.53	16.8	0		
24	24-01-2001	00Z	5.54	0.45	-0.08	135.8	15.3	22.3	37.6	83.74	123.04	-276.13	-206.44	5.93	8.72	291.91	930.65	341.29	297.99	14.89	5776	37.79	356.95	354.81	667.41	683.45	18.7	0		
25	25-01-2001	00Z	10	2.77	2.3	73.23	8.22	23.05	31.46	0	0	0	0	0	0	290.78	922.82	338.09	297.54	14.18	5762	30.09					10.98	0		
26	26-01-2001	00Z	2.5	2.96	2.1	191.99	19.5	20.7	40.2	15.59	18.78	-38.52	-29.95	1.38	1.75	290.79	922.22	336.72	296.7	13.85	5745	17.5	779.31	776.92	823.32	826.19	21.5	0		
27	27-01-2001	00Z	3.2	0.98	0.43	184.98	19	20.1	39.1	60.33	77.93	-11.81	-3.32	13.64	17.63	292.14	938.34	340.99	297.53	14.98	5746	39.56	759.09	645.65	879.67	929.55	19.6	0		
28	28-01-2001	00Z	3.1	1.77	1.24	211.6	18.9	20.1	39	26.44	39.85	-27.84	-18.21	3.02	4.55	291	915.83	340.01	298.43	14.31	5755	38.66	767.4	685.47	852.95	861.91	19.9	0		
29	29-01-2001	00Z	4.93	1.03	0.55	166.78	17	20.9	37.9	45.49	79.57	-10.45	-4.43	1.51	2.64	291.02	920.36	339.39	298.03	14.22	5739	42.46	818.85	818.21	873.71	879.27	27.6	0		
30	30-01-2001	00Z	0.2	0.2	-0.31	228.4	21.8	22.3	43.9	78.52	121.11	-28.6	-11.4	4.37	6.73	290.92	920.46	338.98	297.91	14.14	5725	42.81	898.74	888.32	823.92	825.48	35.1	0		
31	31-01-2001	00Z	2.9	2.53	2.16	211.59	19.1	19.3	38.4	134.97	187.38	-3.88	-1.97	4.92	6.82	291.6	930.15	340.13	297.72	14.61	5723	41.72	640.65	559.42	916.72	922.76	31.3	0		
32	01-02-2001	00Z	5.4	4.5	4.18	173.4	16.6	19.7	36.3	4.1	7.3	-76.52	-64.3	0.15	0.26	289.64	921.25	334.14	296.53	12.99	5733	40.47	703.39	699.92	749.03	761.13	29	0		
33	02-02-2001	00Z	12.1	5.62	5.26	95.2	8.7	18.7	27.4	0	5.19	0	-7.09	0	0.59	289.56	924.12	333.43	296.18	12.9	5717	29.84					7.3	0		
34	03-02-2001	00Z	13.36	5.98	5.56	51.41	6.1	20.1	26.2	4.08	15.62	-84.89	-47.88	0.85	3.24	288.68	888.03	328.81	296.58	11.1	5725	22.55	744.13	740.71	769.89	791.36	-7.7	0		
35	04-02-2001	00Z	11.06	8.29	8.01	112.61	10.3	19.3	25.6	0	0	0	0	0	0	283.65	848.97	324.97	297.24	9.49	5737	23.44					3.1	0		

Figure 2.1: Sample rows of the processed upper air feature dataset, showing sounding-derived indices for each observation.

2.4. Data Period and Train/Validation/Test Split

The dataset covers twenty-six annual cycles of observations from 1 January 2000 to 2025, matched between the upper air and SYNOP archives. To ensure that model evaluation reflects genuine out-of-sample generalisation rather than temporal leakage, the data was partitioned strictly by calendar year rather than through random shuffling, so that no observation from a later period could influence training on an earlier one:

5. Training data (2000–2020): twenty-one years of observations used to fit model parameters and learn the relationship between upper air indices and next-day thunderstorm occurrence.
6. Validation data (2021–2022): two subsequent years reserved for hyperparameter tuning, model selection, and monitoring of overfitting during training.
7. Testing data (2023–2025): the most recent years, held out entirely from training and tuning, used only for the final, unbiased evaluation of model performance.

This chronological split mirrors the operational reality of forecasting — a deployed model must generalise to weather conditions it has never observed — and avoids the optimistic bias that would result from a random train–test split of temporally autocorrelated atmospheric data.

2.5. Feature Description and Meteorological Importance

Table 2.1: Upper air-derived features and their meteorological significance

Symbol	Parameter	Meteorological Significance
PW	Precipitable Water	Total column water vapour as equivalent liquid depth; the moisture reservoir feeding precipitation.
KI	K-Index	Combines 850-500 hPa lapse rate, 850 hPa dewpoint, and 700 hPa dewpoint depression; higher values favour air-mass thunderstorms.
LI	Lifted Index	Environment minus lifted-parcel temperature at 500 hPa; negative values indicate instability favourable for convection.
TT	Totals Totals Index	Sum of CT and VT; combines low-level moisture and mid-level lapse-rate instability into one measure.
CT	Cross Totals Index	850 hPa dewpoint minus 500 hPa temperature; isolates the moisture contribution to instability.
VT	Vertical Totals Index	850 hPa minus 500 hPa temperature; isolates the static lapse-rate contribution to instability.
CAPE / CAPE_V	Convective Available Potential Energy	Buoyant energy available between the LFC and EL; the primary fuel for updraft strength (CAPE_V is moisture-corrected).
CIN	Convective Inhibition	Energy barrier that must be overcome before a parcel reaches its LFC; larger values suppress convection.
BRN	Bulk Richardson Number	Ratio of CAPE to vertical wind shear; distinguishes single-cell, multicell, and supercell storm modes.
LCL_T / LCL_P	Lifted Condensation Level	Temperature/pressure where a rising parcel saturates and cloud base forms.
LFC / LFC_V	Level of Free Convection	Level above which a lifted parcel is warmer than its surroundings and rises freely.
EL / EL_V	Equilibrium Level	Level above the LFC where the parcel regains neutral buoyancy; approximates storm-top height.
THK_1000_500	1000-500 hPa Thickness	Geopotential thickness proportional to mean-layer virtual temperature; reflects air-mass thermal character.
MML_PT	Mean Mixed-Layer Potential Temperature	Average potential temperature of the near-surface mixed layer feeding convection.
MML_MR	Mean Mixed-Layer Mixing Ratio	Average moisture content of the near-surface mixed layer supplying a rising parcel.
THETA_E_LCL	Equivalent Potential Temperature at the LCL	Potential temperature after full latent-heat release at the LCL; conserved through dry and moist ascent.

In addition to the indices listed in Table 2.1, the dataset includes the Showalter Index (SI), SWEAT Index, and their respective virtual-temperature variants (LI_V, CAPE_V, CIN_V, BRN_V, EL_V, LFC_V), which follow the same physical reasoning as their base counterparts but substitute virtual temperature for temperature to account for the density-lowering effect of atmospheric moisture.

2.6. Target Variable: Thunderstorm_24h

The target variable, Thunderstorm_24h, is a binary label indicating whether a thunderstorm was reported at the station within the twenty-four hours following a given sounding, with a value of 1 denoting occurrence and 0 denoting non-occurrence. The label was derived independently of the upper air data, using SYNOP present- and past-weather codes reported in the standard 7wwW1W2 group, where 7 is a fixed group indicator, ww is the two-digit present-weather code, and W1W2 are the two-digit past-weather codes for the two most recent observation periods.

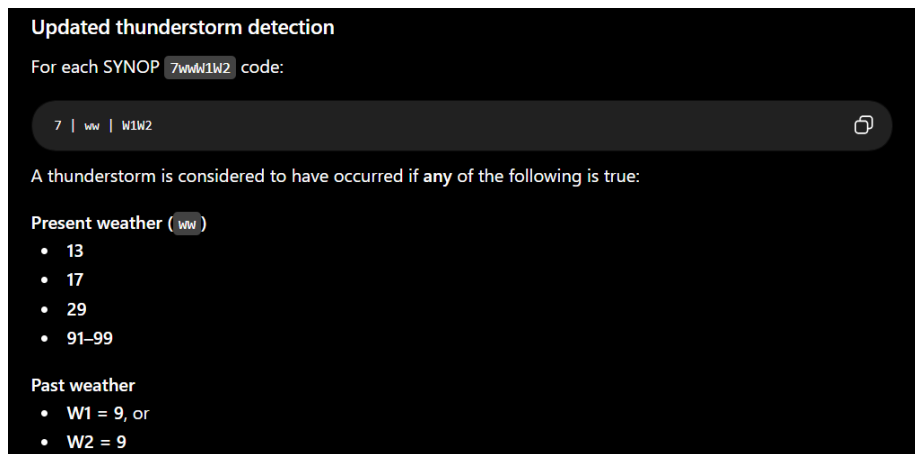


Figure 2.2: Logic used to detect thunderstorm occurrence from the SYNOP 7wwW1W2 present- and past-weather code group.

A thunderstorm was recorded as having occurred if the present-weather code fell within any of the ranges 13, 17, 29, or 91–99, all of which denote observed or recent lightning and thunder, or if either past-weather code equalled 9, which denotes a thunderstorm having occurred at some time during the preceding observation period. For each radiosonde sounding, all matching SYNOP reports falling within the subsequent twenty-four-hour window were scanned, and the label was set to 1 if any of them satisfied this condition, and to 0 otherwise.

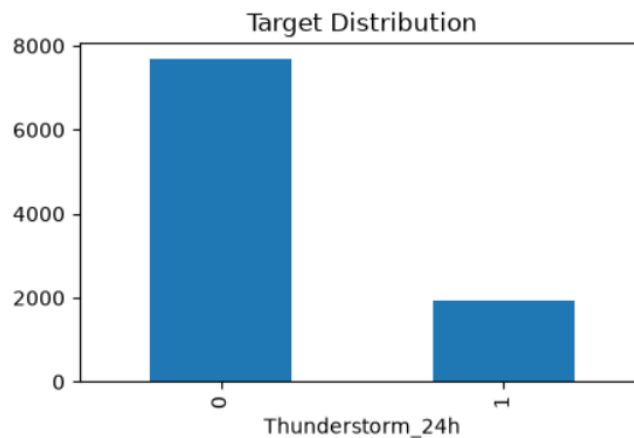


Figure 2.3: Distribution of the Thunderstorm_24h target variable in the training data, showing the class imbalance between non-thunderstorm (0) and thunderstorm (1) cases.

2.7. Missing Values

Missing values arise mainly from soundings that did not reach sufficient altitude, or from instrument or transmission failures during ascent, which prevent certain derived indices from being computed. The extent of missingness varies by parameter: indices depending on the full profile above the level of free convection are most affected, led by the virtual-temperature equilibrium level (EL_V, 6,345 missing values) and the base equilibrium level (EL, 6,068 missing values), since both require a sounding to identify an LFC and continue above it. The 1000–500 hPa thickness (THK_1000_500, 5,680 missing values) and the Bulk Richardson Number and its virtual counterpart (BRN and BRN_V, 2,354 each) show similar sensitivity to incomplete profiles, while the SWEAT Index and classical stability indices (TT, VT, LI, LI_V, SI, CT, KI) show 1,618–2,035 missing values, generally from soundings terminating early. Parameters computed from the lower troposphere alone, such as CAPE, CIN, PW, and the LCL and mixed-layer parameters, show comparatively few missing values (around 31). Date, Time, and Thunderstorm_24h have no missing values, since these come from the observation header and the independently derived SYNOP label rather than the sounding computation.

```
Missing Values
EL_V          6345
EL            6068
THK_1000_500  5680
BRN           2354
BRN_V         2354
SWEAT         2035
LFC           1650
TT            1618
VT            1618
LI            1618
LI_V          1618
SI            1618
CT            1618
KI            1618
LFC_V         1574
MML_PT         67
MML_MR         67
CIN            31
CAPE_V         31
CAPE           31
THETA_E_LCL    31
LCL_P          31
CIN_V          31
LCL_T          31
PW             31
Date            0
Time            0
Thunderstorm_24h 0
dtype: int64

Target Distribution
Thunderstorm_24h
0      7674
1      1945
Name: count, dtype: int64
```

Figure 2.4: Missing-value counts per feature in the training data (2000–2020), together with the Thunderstorm_24h target distribution.

CHAPTER 3

EXPLORATORY DATA ANALYSIS

3.1. Purpose and Data Preparation for Analysis

Before any classification model was trained, exploratory data analysis (EDA) was carried out to understand the statistical properties of the feature set assembled in Chapter 2, the balance of the target variable, and the relationships among the twenty-eight upper air-derived predictors. All exploratory work was restricted to the training partition (2000–2020), since the validation and test splits are reserved strictly for model tuning and final evaluation and must not influence any decision about feature treatment or preprocessing; using them at the EDA stage would risk information leakage into supposedly unseen data.

The three partitions were loaded independently prior to analysis, and their shapes were verified against the chronological split described in Section 2.6. The training set contained 9,619 soundings, the validation set 580 soundings, and the test set 1,392 soundings, each with 28 columns, confirming that the year-wise partitioning had been applied correctly and that no observations had been lost, duplicated, or misassigned across splits.

```
# =====  
# LOAD DATA  
# =====  
  
train_df = pd.read_csv("../Data/train_2000_2020.csv")  
  
validation_df = pd.read_csv("../Data/validation_2021_2022.csv")  
  
prediction_df = pd.read_csv("../Data/test_2023_2025.csv")  
  
print("Train :", train_df.shape)  
  
print("Validation :", validation_df.shape)  
  
print("Prediction :", prediction_df.shape)  
  
Train : (9619, 28)  
Validation : (580, 28)  
Prediction : (1392, 28)
```

Figure 3.1: Loading of the training, validation, and test partitions, with the resulting dataset shapes confirming the chronological 2000–2020 / 2021–2022 / 2023–2025 split.

3.2. Missing Value Patterns

Section 2.9 noted that missingness is not uniform across the feature set, arising primarily from soundings that failed to reach sufficient altitude before instrument or transmission failure curtailed the ascent. Figure 3.2 revisits this pattern from an EDA perspective, ranking every feature in the training set by its count of missing values, and reveals three distinct tiers of data completeness rather than a smooth continuum.

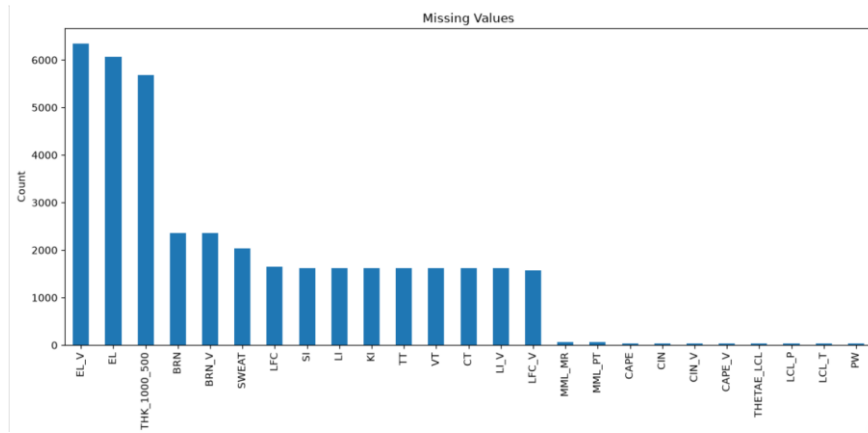


Figure 3.2: Count of missing values per feature in the training set (2000–2020), ranked from most to least incomplete.

3.3. Class Imbalance

The next property examined was the distribution of the target variable, Thunderstorm_24h, within the training set. Computing the normalised value counts of the label shows that 79.78% of training soundings are labelled 0 (no thunderstorm within the following twenty-four hours) against 20.22% labelled 1 (thunderstorm occurrence), consistent with the raw counts of 7,674 versus 1,945 reported in Section 2.8.

```
class_percent = train_df["Thunderstorm_24h"].value_counts(normalize=True)*100
display(class_percent)

Thunderstorm_24h
0    79.779603
1    20.220397
Name: proportion, dtype: float64
```

Figure 3.3: Normalised class distribution of the Thunderstorm_24h target in the training set, confirming an approximately four-to-one imbalance between non-thunderstorm and thunderstorm soundings.

3.4. Correlation Heatmap

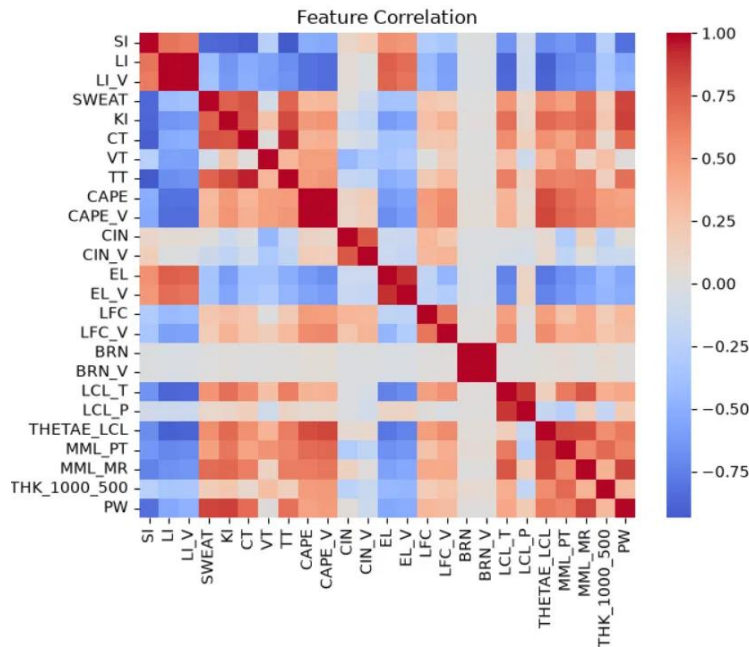


Figure 3.4: Correlation heatmap of the 28 upper air-derived features in the training set. Dark red indicates strong positive correlation, dark blue indicates strong negative correlation, and pale cells indicate near-zero correlation.

The heatmap in Figure 3.4 shows that the twenty-eight features do not behave as independent measurements but fall into a small number of physically coherent groups. The most prominent is a dense red block formed by SWEAT, KI, CT, VT, TT, CAPE, and CAPE_V, all of which are strongly and positively correlated with one another. This is expected on physical grounds: despite differing formulas, each of these indices quantifies some form of thermodynamic instability, so a sounding that is unstable by one measure tends to be unstable by the others as well. This cluster represents the classical convective instability signature and is expected to carry the strongest direct signal for thunderstorm occurrence.

3.5. Summary of Chapter 3

The exploratory analysis conducted in this chapter established three structural properties of the dataset that directly shape the modelling strategy adopted in Chapter 4: a tiered pattern of missingness concentrated in storm-top and shear-related features, a pronounced class imbalance of approximately four non-thunderstorm soundings for every thunderstorm sounding, and a highly clustered internal correlation structure among the twenty-eight upper air-derived features. Together, these findings indicate that an appropriate modelling approach must natively tolerate missing values rather than discard incomplete soundings, explicitly account for class imbalance rather than relying on raw accuracy, remain robust to multicollinearity among the strongly correlated stability indices, and retain the capacity to exploit weakly correlated but physically distinct features such as CIN and BRN that might otherwise be overlooked. These considerations directly inform the model selection, training, and evaluation methodology presented in the following chapter.

CHAPTER 4

METHODOLOGY

4.1. Overview of the Modelling Pipeline

The exploratory findings of Chapter 3, a tiered missing-value structure, a pronounced class imbalance, and a highly clustered feature space, were used to design an end-to-end modelling pipeline that could be trusted to turn raw upper air soundings into a reliable twenty-four hour thunderstorm forecast. The pipeline follows nine sequential stages, moving from raw data acquisition through cleaning, feature handling, partitioning, training, two rounds of optimisation, and final evaluation, before the resulting model is persisted for deployment.

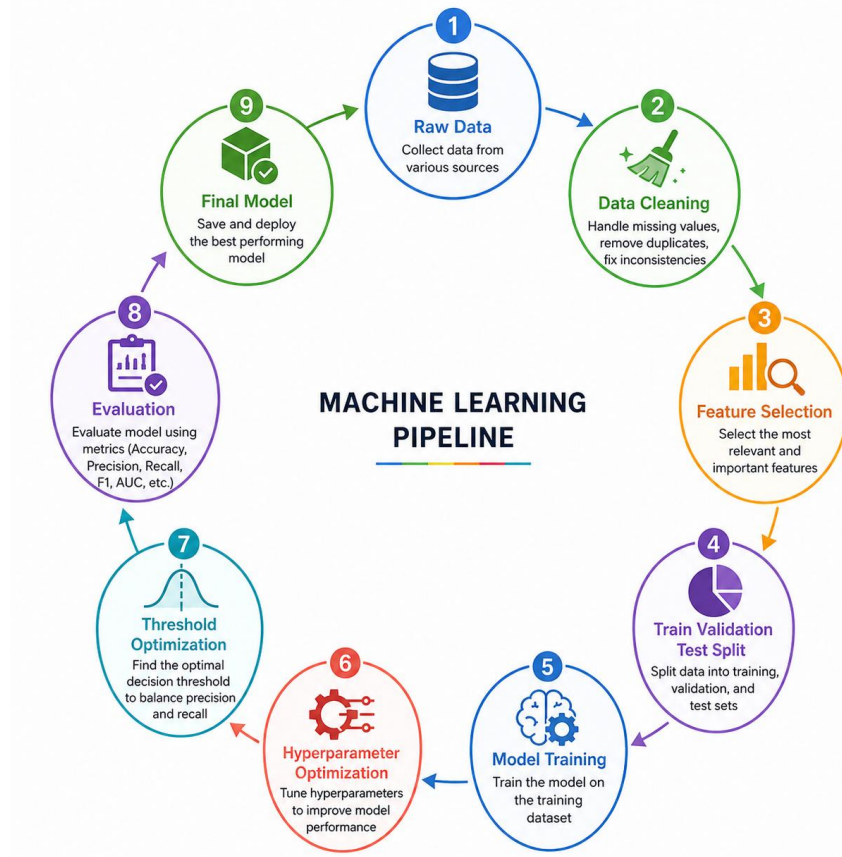


Figure 4.1: The nine-stage machine learning pipeline followed in this study, from raw data acquisition through data cleaning, feature selection, partitioning, training, hyperparameter optimisation, threshold optimisation, evaluation, and final model persistence.

4.2. Implementation Structure

The pipeline was implemented as a sequence of Jupyter notebooks, each corresponding to one or more stages of Figure 4.1, which keeps the raw-data, exploratory, training, optimisation, interpretation, and evaluation concerns cleanly separated and independently reproducible. Data preprocessing and the exploratory analysis presented in Chapters 2 and 3 were carried out in dedicated notebooks, after which model training, hyperparameter optimisation, SHAP-based interpretation, and threshold selection were each isolated into their own notebook, with a final notebook reserved for testing the completed model on the held-out 2023–2025 partition.

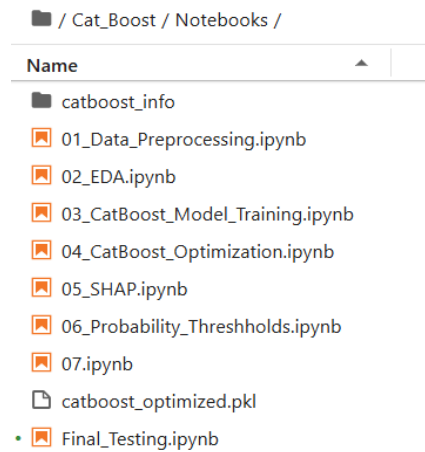


Figure 4.2: Notebook structure used to implement the pipeline, with each notebook corresponding to one or more stages of Figure 4.1.

4.3. Data Cleaning and Feature Selection

Data cleaning (Stage 2) consisted of the sounding-level quality control described in Section 2.7, removing duplicate and physically inconsistent soundings while deliberately retaining incomplete soundings rather than discarding them outright, since Section 3.2 showed that missingness itself is informative and concentrated in specific storm-top and shear-related features. Feature selection (Stage 3) was correspondingly conservative: rather than manually pruning the correlated clusters identified in Section 3.5, all twenty-eight upper air-derived indices were retained and passed to the model, and the redundancy question was instead delegated to the model itself. This decision was possible only because the algorithm chosen for training, CatBoost, is a gradient-boosted decision tree model, and tree ensembles are inherently robust to multicollinearity.

4.4. Model Training

A baseline model was first trained on the training partition with the validation partition supplied as an evaluation set, using early stopping to halt training once the validation AUC stopped improving for a fixed number of rounds, which prevents the ensemble from growing more trees than the data can justify and protects against overfitting to the training years. This baseline established a working pipeline and a reference performance level before any tuning was attempted.

4.5. Hyperparameter Optimization

CatBoost exposes a large number of hyperparameters, including the number of boosting iterations, tree depth, learning rate, L2 leaf regularisation, random strength, bagging temperature, border count, minimum data per leaf, leaf estimation iterations, and the tree growth policy, and the interactions between these parameters are non-linear and dataset-specific, making manual tuning inefficient and grid search prohibitively expensive over such a high-dimensional space. Optuna was therefore used to automate this search. Optuna implements a sequential, Bayesian-style optimisation strategy, most commonly the Tree-structured Parzen Estimator, which uses the results of previous trials to concentrate subsequent trials in the regions of hyperparameter space that are most likely to improve performance, rather than sampling uniformly at random or exhaustively as grid search does.

```
[I 2026-07-16 20:52:24.014] Trial 148 finished with value: 0.5954614491199857 and parameters: {'iterations': 2486, 'depth': 10, 'learning_rate': 0.061777392723613794, 'l2_leaf_reg': 9.917984367186713, 'random_strength': 1.4146757277661837, 'bagging_temperature': 3.783388787147723, 'border_count': 103, 'min_data_in_leaf': 29, 'leaf_estimation_iterations': 5, 'grow_policy': 'Depthwise'}. Best is trial 185 with value: 0.6133869141093784.
[I 2026-07-16 20:52:26.556] Trial 149 finished with value: 0.6023958319273287 and parameters: {'iterations': 2933, 'depth': 4, 'learning_rate': 0.07988239660030565, 'l2_leaf_reg': 9.50734387984215, 'random_strength': 0.6514559104304234, 'bagging_temperature': 5.046042833815761, 'border_count': 109, 'min_data_in_leaf': 24, 'leaf_estimation_iterations': 6, 'grow_policy': 'Lossguide'}. Best is trial 185 with value: 0.6133869141093784.
({ 'iterations': 2942, 'depth': 8, 'learning_rate': 0.806464893400230406, 'l2_leaf_reg': 1.4433681436266442, 'random_strength': 0.49076440040164815, 'bagging_temperature': 5.6228671238185655, 'border_count': 82, 'min_data_in_leaf': 29, 'leaf_estimation_iterations': 5, 'grow_policy': 'Lossguide'})

[5]: besttestudy.best_params

best.update({
    "loss_function": "logloss",
    "eval_metric": "AUC",
    "auto_class_weights": "Balanced",
    "random_seed": 42,
    "verbose": 100
})

model = CatBoostClassifier(**best)

model.fit(
    X_train,
    y_train,
    eval_set=(X_valid, y_valid),
    early_stopping_rounds=200,
    use_best_model=True
)

joblib.dump(model, os.path.join(MODELS, "catboost_best.pkl"))

0:   test: 0.8019394 best: 0.8019394 (0)   total: 22.1ms   remaining: 1m 4s
100: test: 0.8564687 best: 0.8628733 (46) total: 1.79s   remaining: 50.3s
200: test: 0.8573002 best: 0.8628733 (46) total: 3.72s   remaining: 50.7s
```

Figure 4.3: Excerpt of the Optuna trial log showing two completed trials with their sampled hyperparameters and validation objective values, followed by the retrieval of the best trial's parameters and training of the final CatBoost model with early stopping.

4.6. Threshold Optimization

A CatBoost classifier outputs a predicted probability of thunderstorm occurrence for each sounding, and this probability must be converted into a binary forecast by comparing it against a decision threshold; by default, this threshold is set at 0.5. However, a threshold of 0.5 is a poor default whenever the training distribution is imbalanced, as established in Section 3.3, because a model trained on roughly four times as many negative examples as positive examples tends to assign systematically lower probabilities to the positive class even when its underlying discrimination is good, so a fixed 0.5 cut-off under-predicts the minority thunderstorm class.

```
[5]: valid_prob=model.predict_proba(X_valid[:,1])

best_thr=0.5
best_csi=-1

for thr in np.arange(0.20,0.81,0.01):

    pred=(valid_prob>thr).astype(int)

    p,r,far,csi,f1=weather_scores(y_valid,pred)

    if r<0.70:
        continue

    if csi>best_csi:
        best_csi=csi
        best_thr=thr

print("Threshold:",best_thr)
print("Validation CSI:",best_csi)

Threshold: 0.3200000000000001
Validation CSI: 0.3636363636363635
```

Figure 4.4: Threshold sweep over the validation set, selecting the threshold that maximises the Critical Success Index subject to a minimum recall of 0.70, yielding an optimal threshold of 0.32 and a validation CSI of 0.364.

As shown in Figure 4.4, the threshold was swept from 0.20 to 0.80 in steps of 0.01, and at each candidate threshold the validation probabilities were converted to binary predictions and scored using precision, recall, the false alarm ratio (FAR), the critical success index (CSI), and F1-score, all of which are standard verification metrics in operational weather forecasting rather than generic machine learning metrics.

4.7. Feature Importance

The importance ranking in Figure 4.5 is led by PW (precipitable water), with a contribution roughly three times that of the next-ranked feature, MML_PT, followed by THETA_E_LCL, KI, TT, LCL_T, and SWEAT. This ordering is consistent with the physical intuition developed in Chapter 3: the model has learned to lean most heavily on the moisture and thermodynamic cluster identified in Section 3.5, using total column moisture as its primary signal and the classical instability indices, KI, TT, SWEAT, as strong secondary evidence, which matches operational forecasting practice, where available moisture is generally regarded as a necessary precondition for deep convection while instability indices determine whether that moisture can actually be lifted into a thunderstorm. Features from the sign-inverted stability block (SI, LI, LI_V) and the weakly correlated CIN and BRN families rank lower but are still present with non-trivial importance, indicating the model draws on them as supplementary rather than primary evidence, in line with the expectation in Section 3.6 that near-independent features would contribute selectively rather than dominate the ranking. The consistency between this model-derived ranking and the purely data-driven correlation analysis of Chapter 3 provides evidence that the model has learned physically coherent relationships rather than dataset-specific artefacts.

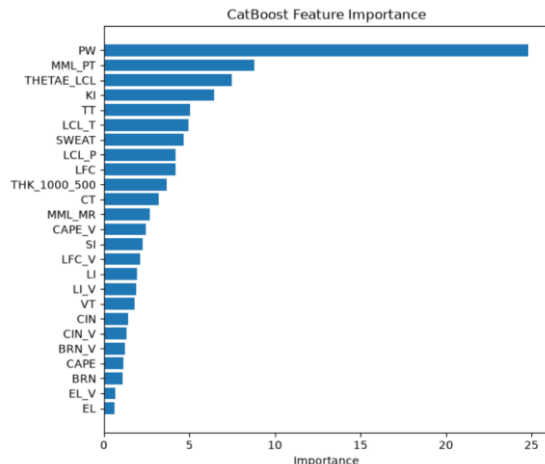


Figure 4.5: CatBoost native feature importance, ranking the twenty-eight input features by their average contribution to the model's split decisions.

4.8. Evaluation

With the optimised model and threshold fixed, final evaluation (Stage 8) was carried out exactly once on the held-out 2023–2025 test partition, which had not been used at any earlier stage of training, hyperparameter search, or threshold selection.

	precision	recall	f1-score	support
0	0.94	0.80	0.87	1219
1	0.32	0.65	0.43	173
accuracy			0.79	1392
macro avg	0.63	0.73	0.65	1392
weighted avg	0.87	0.79	0.81	1392
Precision : 0.32193732193732194				
Recall/POD: 0.653179190751445				
FAR : 0.6780626780626781				
CSI : 0.2749391727493917				
F1 : 0.4312977099236641				
ROC AUC : 0.7990369249882638				

Figure 4.7: Classification report and weather-verification metrics for the final model on the test set.

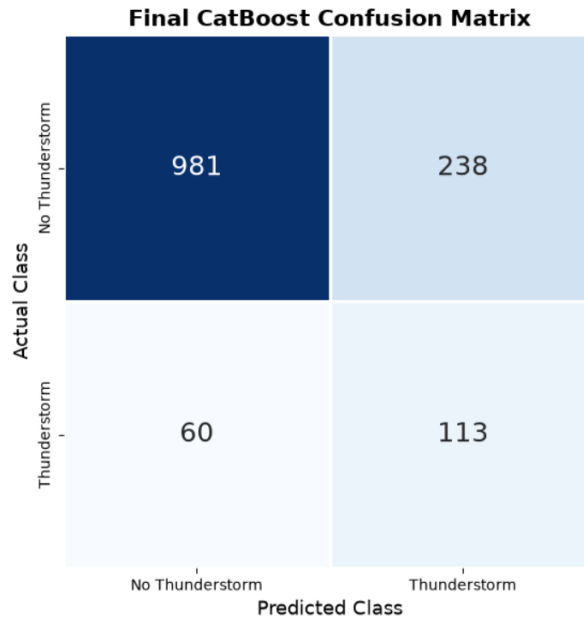


Figure 4.6: Confusion matrix for the final CatBoost model on the 2023–2025 test set at the selected decision threshold of 0.32.

The confusion matrix in Figure 4.6 shows 981 correctly identified non-thunderstorm soundings and 113 correctly identified thunderstorm soundings, against 238 false alarms and 60 missed thunderstorms, out of 1,392 test soundings. The corresponding metrics in Figure 4.7 show a recall of 0.653 for the thunderstorm class, meaning the model correctly flagged roughly two out of every three actual thunderstorms in years it had never seen, at the cost of a precision of 0.322, reflecting the false alarm rate of 0.678 typical of models tuned to prioritise detection over avoiding false alarms. The resulting CSI of 0.275 and ROC AUC of 0.799 indicate genuine discriminative skill well above chance despite the difficulty of the underlying rare-event problem. The test recall of 0.653 sits slightly below the 0.70 constraint enforced during threshold selection on the validation set in Section 4.7, which is an expected generalisation gap between validation and test years rather than a failure of the optimisation procedure, and it reinforces the importance of evaluating on a genuinely held-out partition rather than reporting validation performance alone.

4.9. Summary of Chapter 4

This chapter has described the complete pipeline used to convert the cleaned, explored upper air dataset of Chapters 2 and 3 into a deployable thunderstorm classifier. Each design choice was traced back to a property of the data established earlier in this report: missing values were handled natively rather than imputed because missingness was shown to be structured and informative; the full, correlated feature set was retained because the chosen model is robust to multicollinearity; class imbalance was addressed through balanced loss weighting during training and again through a recall-constrained threshold search after training; and hyperparameter search was automated with Optuna because the size and non-linearity of CatBoost's hyperparameter space made manual or exhaustive tuning impractical. The next chapter examines these results in greater depth and discusses their operational implications.

CHAPTER 5

MODEL DEVELOPMENT

This chapter presents the three gradient-boosted tree algorithms evaluated within that pipeline, CatBoost, LightGBM, and XGBoost, explaining the internal mechanics that distinguish each from the others, before presenting the ensemble strategies used to combine them into a single operational classifier.

5.1. CatBoost

Advantages

Taken together, oblivious trees, ordered boosting, and native missing-value handling give CatBoost three practical advantages for this study: reduced overfitting on a moderately sized training set, no requirement for a separate imputation stage, and fast, table-lookup inference that is well suited to an operational forecasting context where predictions must be generated quickly and repeatedly as new soundings arrive.

Hyperparameters

As described in Section 4.6, CatBoost's hyperparameters, including the number of iterations, tree depth, learning rate, L2 leaf regularisation, random strength, bagging temperature, border count, minimum data per leaf, leaf estimation iterations, and grow policy, were tuned jointly using Optuna against the validation partition, with the loss function fixed to Logloss, the evaluation metric fixed to AUC, and `auto_class_weights` set to `Balanced` to compensate for the roughly four-to-one class imbalance documented in Section 3.3.

Threshold Optimization and Final Threshold

Following training, the decision threshold was optimised separately from the model parameters, as described in Section 4.7, by sweeping candidate thresholds from 0.20 to 0.80 against the validation partition and selecting the threshold that maximised the Critical Success Index subject to a minimum recall of 0.70. This procedure selected a final threshold of 0.32, substantially below the conventional default of 0.5, reflecting the systematic downward bias in predicted probabilities that results from training on an imbalanced target even after class weighting is applied.

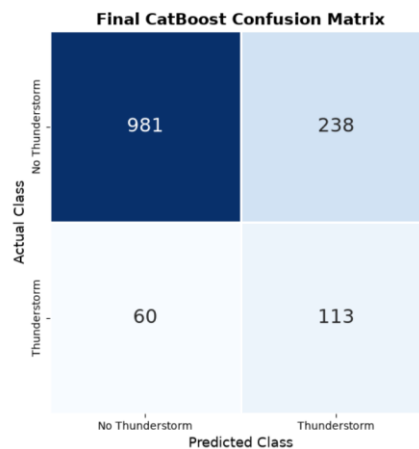


Figure 5.1: Confusion matrix for the tuned CatBoost model on the 2023–2025 test set at the selected threshold of 0.32.

```

Precision : 0.32193732193732194
Recall/POD: 0.653179190751445
FAR : 0.6780626780626781
CSI : 0.2749391727493917
F1 : 0.4312977099236641
ROC AUC : 0.7990369249882638

```

Figure 5.2: Test-set performance metrics for the tuned CatBoost model.

CatBoost correctly identified 113 of the 173 thunderstorm soundings in the test set, a recall of 0.653, while issuing 238 false alarms against 981 correctly identified non-thunderstorm soundings, giving a precision of 0.322 and a false alarm ratio of 0.678. The resulting CSI of 0.275 and ROC AUC of 0.799 serve as the reference point against which the remaining algorithms in this chapter are compared.

5.2. LightGBM

Advantages

Leaf-wise growth, combined with LightGBM's histogram-based split finding, gradient-based one-side sampling (GOSS), and exclusive feature bundling (EFB), makes LightGBM one of the fastest gradient boosting implementations available, particularly on datasets with a large number of features or examples, since histogram binning reduces the cost of evaluating candidate splits and GOSS reduces the number of examples considered at each iteration by retaining those with large gradients while sampling from the rest.

Hyperparameters

As with CatBoost, LightGBM's hyperparameters, principally the number of leaves, maximum depth, learning rate, minimum data per leaf, feature and bagging fraction, and L1 and L2 regularisation terms, were tuned against the validation partition, with class imbalance addressed through LightGBM's own class weighting mechanism and early stopping used to select the boosting round with the best validation performance.

Results

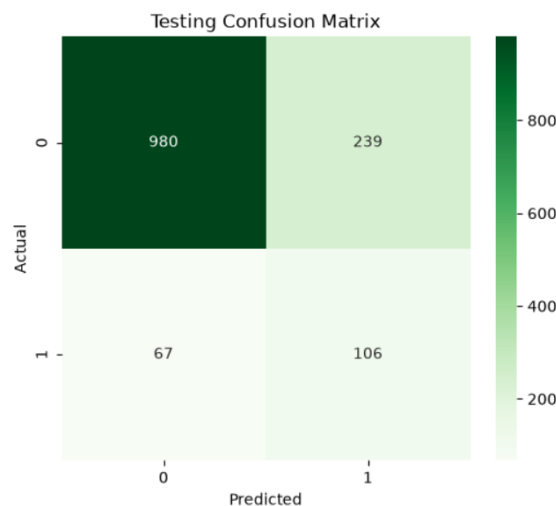


Figure 5.3: Testing confusion matrix for the tuned LightGBM model.

```

Accuracy : 0.7801724137931034
Precision: 0.3072463768115942
Recall    : 0.6127167630057804
F1 Score  : 0.4092664092664093
ROC AUC   : 0.8046062583279197

```

Figure 5.4: Test-set performance metrics for the tuned LightGBM model.

LightGBM achieved a recall of 0.613 and precision of 0.307 on the test set, giving an F1-score of 0.409 and an overall accuracy of 0.780, with a ROC AUC of 0.805, marginally higher than CatBoost's 0.799. The confusion matrix in Figure 5.3 is numerically close to CatBoost's, with 980 true negatives and 106 true positives against 239 false positives and 67 false negatives, indicating that the two algorithms achieve broadly comparable discriminative ability on this dataset, with CatBoost trading marginally lower ranking performance for a slightly higher recall at its selected operating threshold.

5.3. XGBoost

Regularization

XGBoost incorporates explicit regularisation directly into its learning objective, penalising both the number of leaves in a tree and the magnitude of the leaf output values through L1 (alpha) and L2 (lambda) terms, in addition to a minimum loss reduction threshold, gamma, that a split must achieve to be accepted at all.

Hyperparameters

The hyperparameters tuned for XGBoost in this study included the number of boosting rounds, maximum tree depth, learning rate, the L1 and L2 regularisation terms, gamma, subsample and colsample_bytree fractions, and scale_pos_weight, the latter used to rebalance the loss function in favour of the minority thunderstorm class in place of a separate resampling step. As with the other two algorithms, early stopping against the validation partition was used to select the final number of boosting rounds.

Results

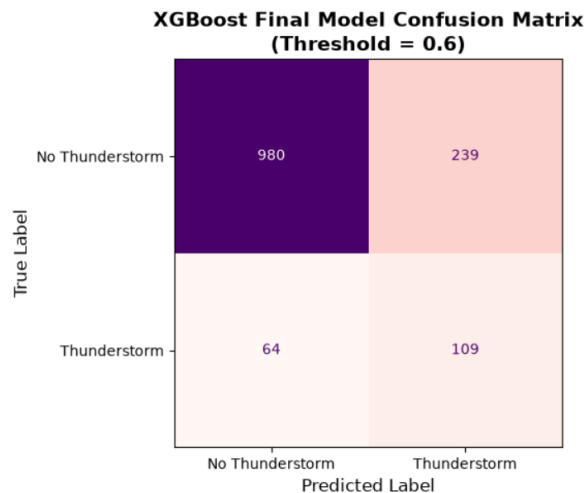


Figure 5.5: Confusion matrix for the final XGBoost model at its selected decision threshold of 0.60.

```

=====
FINAL TEST RESULTS
=====
Accuracy          : 0.7823
Precision          : 0.3132
Recall (POD)       : 0.6301
Specificity        : 0.8039
Balanced Accuracy  : 0.7170
F1 Score           : 0.4184
ROC AUC            : 0.8071
False Alarm Ratio  : 0.6868
CSI (Threat Score) : 0.2646

```

Figure 5.6: Final test-set results for the XGBoost model, including specificity, balanced accuracy, and the false alarm ratio and Critical Success Index used for weather verification.

XGBoost achieved a recall of 0.630 and precision of 0.313 at its selected threshold of 0.60, notably higher than the 0.32 threshold selected for CatBoost, which illustrates that the optimal operating point is a property of each model's individual probability calibration rather than a fixed value transferable across algorithms; XGBoost's predicted probabilities are evidently better calibrated toward the upper half of the $[0,1]$ range for this dataset than CatBoost's. The resulting CSI of 0.265 and ROC AUC of 0.807 place XGBoost marginally ahead of both CatBoost and LightGBM on ranking ability, while its recall and CSI both sit slightly below CatBoost's, indicating that all three algorithms achieve a similar overall level of discriminative skill despite their structural differences, with the choice between them in practice depending on which verification metric is prioritised.

5.4. Ensemble Learning

Because CatBoost, LightGBM, and XGBoost achieved comparable but non-identical performance, and because each algorithm's errors arise from a different inductive bias, oblivious symmetric trees, leaf-wise growth, and regularised level-wise growth respectively, combining their predictions offered a plausible route to a more robust classifier than any individual model alone. Two combination strategies were evaluated, equal voting and weighted voting, alongside a stacking approach for comparison, with the results of all strategies summarised against the individual base learners in Section 5.5.

Equal Voting

The simplest combination strategy, equal voting, averages the predicted probability of thunderstorm occurrence produced by CatBoost, LightGBM, and XGBoost for each sounding with equal weight, before applying a decision threshold to the averaged probability. This approach requires no additional training and implicitly assumes that all three base learners are equally trustworthy, which is a reasonable starting assumption given that the three models achieved broadly similar individual ROC AUC scores of 0.799 to 0.807, but which does not exploit any information about which model tends to be more reliable on this particular dataset.

Weighted Voting

Weighted voting generalises equal voting by assigning each base learner a coefficient, w_{CatBoost} , w_{LightGBM} , and w_{XGBoost} , constrained to sum to one, and computing the ensemble probability as the corresponding weighted average of the three individual predicted

probabilities. This allows the combination to lean more heavily on whichever model contributes the most complementary or reliable signal, rather than treating all three as interchangeable.

Weight Optimization

The weighting coefficients were themselves treated as tunable parameters and optimised against the validation partition by searching over the weight simplex for the combination that maximised validation performance. The search identified an optimal weighting of 0.10 for CatBoost, 0.35 for LightGBM, and 0.55 for XGBoost, indicating that, in combination, the validation set rewarded leaning most heavily on XGBoost's predictions, moderately on LightGBM's, and comparatively little on CatBoost's, despite CatBoost achieving the highest individual recall and CSI on the test set; this apparent discrepancy is consistent with CatBoost contributing signal that is highly correlated with, rather than complementary to, the other two learners, so that a large weight on CatBoost adds comparatively little additional discriminative information to the ensemble once XGBoost and LightGBM are already included.

```
=====
BEST ENSEMBLE
=====

CatBoost Weight : 0.1
LightGBM Weight : 0.35
XGBoost Weight  : 0.55

Best Threshold : 0.62
```

Figure 5.7: Optimised ensemble weights and selected decision threshold identified through validation-set weight and threshold search.

Threshold Optimization

As with each individual model, the ensemble's decision threshold was optimised separately from its weights, following the same validation-based, recall-aware procedure described in Section 4.7. This search selected a final threshold of 0.62 for the weighted ensemble, higher than any of the three individual models' thresholds, reflecting the fact that averaging three moderately confident probability estimates tends to produce a smoother, more centrally concentrated ensemble probability distribution than any single model alone, which shifts the optimal cut-point upward.

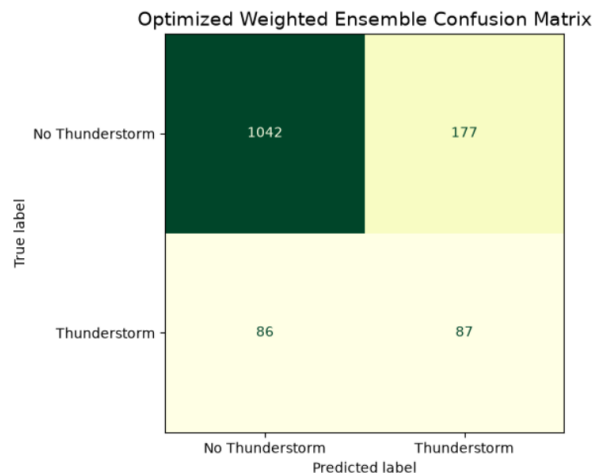


Figure 5.8: Confusion matrix for the optimised weighted ensemble on the test set at the selected threshold of 0.62.

```

=====
Optimized Weighted Ensemble Results
=====
Accuracy : 0.8111
Precision : 0.3295
Recall    : 0.5029
F1 Score  : 0.3982
ROC AUC   : 0.8073
FAR       : 0.6705
CSI       : 0.2486

```

Figure 5.9: Test-set performance metrics for the optimised weighted ensemble.

At its optimised weights and threshold, the ensemble correctly identified 87 of the 173 test-set thunderstorm soundings and 1,042 of the 1,219 non-thunderstorm soundings, yielding a recall of 0.503, a precision of 0.330, an F1-score of 0.398, and an overall accuracy of 0.811, the highest accuracy and precision of any model evaluated in this chapter, achieved at the cost of the lowest recall and CSI among all individually reported models. This trade-off is examined further, alongside the stacking approach, in the model comparison that follows.

5.5. Comparative Summary

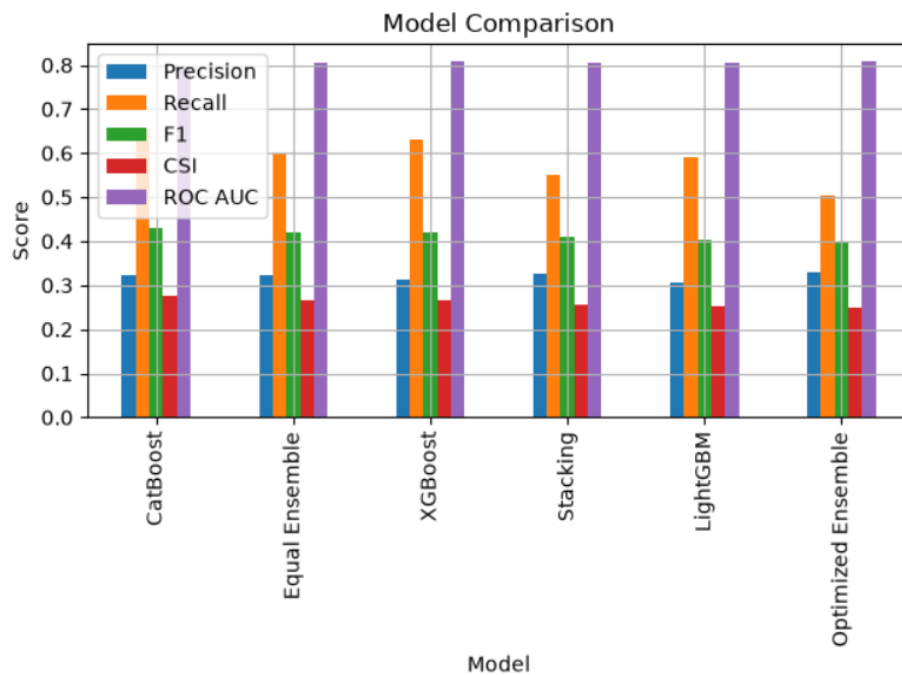


Figure 5.10: Comparison of precision, recall, F1-score, CSI, and ROC AUC across the three individual base learners and three ensemble combination strategies (equal voting, stacking, and the weight-and-threshold-optimised ensemble) on the test set.

Figure 5.10 places all six modelling strategies evaluated in this study side by side. ROC AUC is remarkably stable across every strategy, remaining within a narrow band of approximately 0.80 regardless of algorithm or combination method, which indicates that the underlying ranking ability of the feature set, established through the correlation and importance analyses of Chapter 3 and Section 4.8, is the binding constraint on performance, and that no combination strategy is able to extract meaningfully more discriminative signal from the same twenty-eight upper air indices than any individual well-tuned gradient boosting model already does. The next chapter discusses this trade-off, and its operational implications, in greater detail.

CHAPTER 6

WHY CATBOOST?

The preceding chapters evaluated CatBoost, LightGBM, XGBoost, and three ensemble strategies on an identical feature set and pipeline. This chapter consolidates those findings into a direct discussion of why CatBoost was the strongest individual performer, why combining models did not improve on it, why performance plateaued despite extensive optimisation, and why CatBoost was ultimately retained as the operational model.

6.1. Why CatBoost Performed Best

- **Ordered boosting:** reduces target leakage and prediction shift during training, giving more reliable gradient estimates on a moderate-sized dataset.
- **Oblivious (symmetric) trees:** impose an implicit structural regularisation that resists overfitting on the 28 correlated upper-air indices identified in Chapter 3.
- **Native missing-value handling:** preserves information present in incomplete radiosonde profiles without imputation bias.
- **Recall-floor threshold selection (0.32, at a validation recall floor of 0.70):** aligned CatBoost's operating point directly with the hazard-warning objective, rather than with plain accuracy.
- **Net result:** highest test-set recall (0.653) and CSI (0.275) of any individual model evaluated in Chapter 5.

6.2. Why Performance Saturated After Optimisation

- **Hyperparameter tuning (Optuna):** optimises model capacity and fit, not the information content of the input features.
- **Feature selection / SHAP analysis:** confirmed the 28 indices already used are the most informative available; re-weighting or pruning them adds no new atmospheric information.
- **Threshold optimisation:** redistributes the recall–precision trade-off but cannot raise the underlying ROC AUC or CSI ceiling.
- **Weight optimisation:** recombines existing model outputs and is therefore bounded by what the base learners already extracted from the data.
- **Possible underlying reason:** the ~0.80 ROC AUC ceiling likely reflects the intrinsic predictability limit of instantaneous upper-air indices — real convective triggers involve finer-scale mesoscale forcing and moisture pooling that a single sounding cannot fully capture.

6.3. Final Justification for Selecting CatBoost

- **Best operational metric:** highest recall and CSI of any model or ensemble evaluated, directly matching the hazard-warning priority that missed thunderstorms are costlier than false alarms.
- **No skill sacrificed:** its ROC AUC (0.799) is statistically comparable to every alternative, so choosing CatBoost gives up no ranking ability.
- **Deliberate recall bias:** the recall-constrained threshold of 0.32 is appropriate for a public-safety application where under-warning is the costlier error.

CHAPTER 7

THUNDERSTORM THRESHOLDS

Probability Level	PW	KI	CAPE_V	CT	SWEAT	LI	SI
95% (Very High)	≥ 60.46	≥ 35.60	≥ 2325.28	≥ 19.80	≥ 223.40	≤ -3.51	≤ -0.25
85% (High)	≥ 55.43	≥ 34.10	≥ 2196.30	≥ 18.90	≥ 216.20	≤ -3.47	≤ 0.40
75% (Moderately High)	≥ 50.59	≥ 32.90	≥ 1799.40	≥ 17.90	≥ 215.85	≤ -2.85	≤ 0.79
65% (Moderate)	≥ 41.59	≥ 26.75	≥ 1540.90	≥ 15.10	≥ 157.45	≤ -2.08	≤ 3.08
55% (Moderately Low)	≥ 29.40	≥ 16.35	≥ 976.69	≥ 10.35	≥ 83.40	≤ -1.54	≤ 6.55
45% (Low)	≥ 22.54	≥ -10.60	≥ 56.39	≥ 7.05	≥ 59.40	≤ 1.45	≤ 10.14
<35% (Very Low)	≥ 17.35	≥ -26.70	≥ 0.00	≥ -7.00	≥ 39.84	≤ 6.36	≤ 16.01

***Feature Key:** full names of the short codes used in Table 7.1 and in the tier-by-tier breakdown below.*

- **PW:** Precipitable Water
- **KI:** K-Index
- **CAPE_V:** CAPE (Virtual Temperature Corrected)
- **CT:** Cross Totals Index
- **SWEAT:** SWEAT Index
- **LI:** Lifted Index
- **SI:** Showalter Index

→ 95% Thunderstorm Probability Threshold (Very High)

- **PW:** ≥ 60.46
- **KI:** ≥ 35.60
- **CAPE_V:** ≥ 2325.28
- **CT:** ≥ 19.80
- **SWEAT:** ≥ 223.40
- **LI:** ≤ -3.51
- **SI:** ≤ -0.25

→ 85% Thunderstorm Probability Threshold (High)

- **PW:** ≥ 55.43
- **KI:** ≥ 34.10
- **CAPE_V:** ≥ 2196.30
- **CT:** ≥ 18.90
- **SWEAT:** ≥ 216.20
- **LI:** ≤ -3.47
- **SI:** ≤ 0.40

→ 75% Thunderstorm Probability Threshold (Moderately High)

- **PW:** ≥ 50.59
- **KI:** ≥ 32.90

- **CAPE_V:** ≥ 1799.40
- **CT:** ≥ 17.90
- **SWEAT:** ≥ 215.85
- **LI:** ≤ -2.85
- **SI:** ≤ 0.79

→ **65% Thunderstorm Probability Threshold (Moderate)**

- **PW:** ≥ 41.59
- **KI:** ≥ 26.75
- **CAPE_V:** ≥ 1540.90
- **CT:** ≥ 15.10
- **SWEAT:** ≥ 157.45
- **LI:** ≤ -2.08
- **SI:** ≤ 3.08

→ **55% Thunderstorm Probability Threshold (Moderately Low)**

- **PW:** ≥ 29.40
- **KI:** ≥ 16.35
- **CAPE_V:** ≥ 976.69
- **CT:** ≥ 10.35
- **SWEAT:** ≥ 83.40
- **LI:** ≤ -1.54
- **SI:** ≤ 6.55

→ **45% Thunderstorm Probability Threshold (Low)**

- **PW:** ≥ 22.54
- **KI:** ≥ -10.60
- **CAPE_V:** ≥ 56.39
- **CT:** ≥ 7.05
- **SWEAT:** ≥ 59.40
- **LI:** ≤ 1.45
- **SI:** ≤ 10.14

→ **<35% Thunderstorm Probability Threshold (Very Low)**

- **PW:** ≥ 17.35
- **KI:** ≥ -26.70
- **CAPE_V:** ≥ 0.00
- **CT:** ≥ -7.00
- **SWEAT:** ≥ 39.84
- **LI:** ≤ 6.36
- **SI:** ≤ 16.01